

Where all four frameworks converge — and the main causality axiom becomes a theorem

Bridge Document | DA-1 / T-SNAP Update

ZP Companion | Version 1.7 | April 2026

This companion explains the ideas in plain language with diagrams and real-world examples. It is not the formal document — every claim here restates a result already proved in the corresponding technical document. Consult that document for the authoritative mathematics.

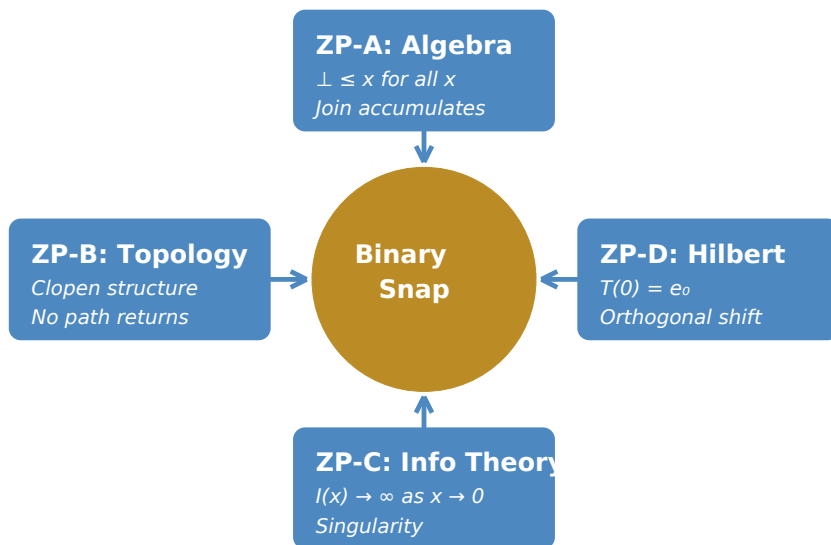
What Is ZP-E Doing?

ZP-E is the cross-framework synthesis. Written last — after ZP-A through ZP-D are each internally closed — its job is to show that the four independent frameworks all describe the same event: the Binary Snap, from four different mathematical vantage points.

ZP-E does not re-derive anything. It imports closed results from each sub-document and shows their consistency. Where a cross-framework connection requires an assumption, that assumption is named explicitly as a bridge axiom — not hidden inside the argument.

The Four Descriptions of the Same Event

The Binary Snap — the transition from nothing ($\perp$) to the first state (ε_0) — looks different depending on which mathematical language you use. ZP-E's central result is that all four descriptions are consistent: one event, four angles.



The Binary Snap (amber) described simultaneously in all four frameworks. Each arrow is an independent proof of the same structural fact.

The Binary Snap (amber center) described simultaneously in all four frameworks. Each arrow represents an independent proof of the same structural fact.

Real-world example — A car crash described by four witnesses

An engineer (forces), a doctor (injuries), a lawyer (liability), and a physicist (energy) each describe the same crash completely within their own discipline. ZP-E shows the four mathematical frameworks are in exactly this relationship to the Binary Snap.

Remember: The car crash illustrates what it means to describe one event in multiple frameworks. The Zero Paradox is not about car crashes — the car crash is an analogy for the forced first transition from $\perp$ to ϵ_0 in any join-semilattice.

The Central Advance: AX-1 is Now a Theorem

In earlier versions, the Binary Snap causality was listed as AX-1 — an axiom: a foundational assumption that could not be derived. The claim was: when P_0 is reached, the Snap happens. Why? Because AX-1 says so.

The DA-1 insert changes this. The argument is now complete: reaching P_0 means a live machine configuration exists (DA-1). Any live configuration passes through c_1 (definition). c_1 is non-null (L-RUN). No program avoids this (TQ-IH). A non-null state change from $\perp$ is the Binary Snap (ZP-A D2). The Snap is derived — not assumed.

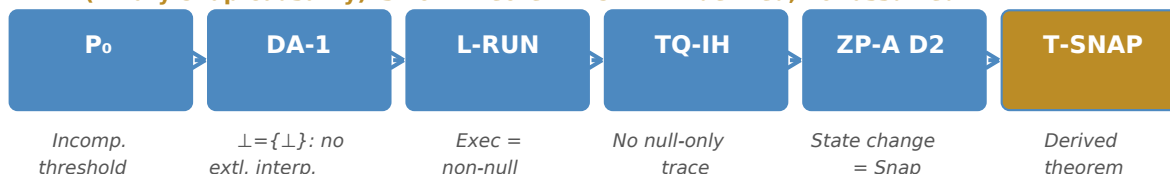
In v2.7, DA-1 itself is upgraded from a Design Principle to a Derived Proposition. Previously DA-1 was an honest but freestanding commitment: "a configuration at P_0 is necessarily executing." Now it follows from ZP-A CC-2: $\perp = \{\perp\}$. The null state is a Quine atom — a self-containing object with no external position from which it could be interpreted as a static description. A thing that interprets itself cannot be waiting for an external interpreter. So $\perp$ at P_0 is necessarily executing. The design commitment has become a derivation.

The two-layer structure of DA-1 (v3.0): DA-1 now rests on two explicit layers. The first is the formal conditional: DP-2 (Execution Distinguishability) establishes that machine states carry execution history independently of output values. From DP-2, Lean 4 can derive `da1_minimal_path` — a proof that before and after instantiation produce the same output value (c_0) while the machine state changes ($c_0 \rightarrow c_1$). `#print axioms` confirms zero axiom dependencies. This is the first Lean formalization of the core DA-1 claim, not just the surrounding algebra. The second layer asks: does $\perp$ actually satisfy DP-2's precondition? That case rests on three converging arguments.

Three paths to the precondition: (1) CC-2/R3 (ZP-A): $\perp = \{\perp\}$ is a Quine atom — it interprets itself, leaving no external position from which it could be read as a static description (above). (2) L-INF (ZP-C): the surprisal of $\perp$ diverges to infinity — no finite static distribution can represent the null state. (3) AIT bridge (v2.8): at the incompressibility threshold P_0 , the description of $\perp$ is maximally incompressible — $K(c_1|n)/|c_1| = 1$. A string that cannot be compressed beyond itself must be its own execution; a static-description reading is ruled out by information theory alone. All three share D7's static/executing dichotomy as background and none is circular with DP-2.

Lean 4 formal closure (ZP-K, v3.7): The three paths are not only conceptually convincing — two of them are now machine-checked. ZP-K adds a `KleeneStructure` instance for `MachinePhase`: it provides a concrete computational Quine (a code that is its own program, via Kleene's second recursion theorem), and proves that this Quine and the AFA self-containment argument are the same structural fact in two different languages. The result is `da1_closed_concrete`: in Lean 4, `IsQuineAtom( $\perp$  : MachinePhase)` is a proved theorem. Path 1 (AFA self-execution) and Path 3 (computational Kleene fixed point) are now formally IN LEAN SCOPE. Path 2 (informational bridge — unbounded surprisal $\rightarrow$ necessarily executing) remains an ontological claim outside Lean scope. The formal grounding of DA-1 is therefore: DP-2 plus two Lean-verified structural paths.

AX-1 (Binary Snap Causality) is now Theorem T-SNAP — derived, not assumed.



The T-SNAP derivation chain: six steps, no axioms beyond AX-B1 and the definition of a Turing machine. AX-1 (amber) is now a theorem.

Real-world example — A legal case that becomes undeniable

A court case starts with an assumption: "the defendant was at the scene." Then surveillance footage, phone records, and witness testimony all confirm it. The assumption becomes a proven fact. T-SNAP is the mathematical equivalent: what was assumed (AX-1) is now proven (T-SNAP) by an independent chain of evidence.

What the Framework Still Assumes

After T-SNAP, the framework rests on exactly three commitments — none of them novel starting assumptions:

Commitment	Statement
AX-B1	Binary Existence. A state either exists or it does not. No third option. Directly verifiable by computation — not a novel commitment.
AX-G1	Initial Object Exists. There is a starting point that reaches everything. Not a novel commitment — grounded in $\perp$ as the bottom element of the ZP-A semilattice.
AX-G2	Source Asymmetry. Nothing returns to the initial object. The origin is unreachable from outside. Not a novel commitment — follows from ZP-A antisymmetry and ZP-B C3.

AX-1 is no longer on this list. The framework makes no stronger claim than it has to.

Remember: The structural results — monotonicity, clopen separation, informational singularity, orthogonal shifts — hold in any instantiation of the framework. The framework makes no claims about which physical theory, if any, instantiates it.